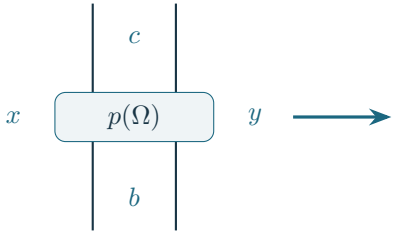
One polynomial for each local operator



The four ordered faces
of one crossing region

$$H_p(x,b,c,y) \;=\; \sum_{i+k+l+t=d} a_p(i,k,l,t)\, b^i x^k y^l c^t$$

$$d \;=\; N-1, \qquad i,k,l,t \;\geq\; 0$$

Coefficients retain phases and derivatives
at repeated spectral roots.

Multiply these polynomials along the original marked diagram.
Every shared face uses one common variable.

$$S_N \;=\; \left[\prod_{f \text{ ordinary}} L_f^{\,d}\right] \prod_{e} H_e \Big|_{L_u = L_v = 0}$$

The original mark gives the two zero targets u, v.

There are r crossing regions and r ordinary faces.
Coefficient extraction enforces every shared face degree.

An exact finite identity in the stated diagram classes, $N \;\geq\; 3$.
The volume theorems also require their analytic and geometric estimates.